

Closed-form metal line-cooling formulas in EXHALE

1. Overview and conventions

All metal line-cooling coefficients in `Cool_coeff.f90` are evaluated from closed-form analytic expressions fitted to the CHIANTI v11.0.2 database (Burgess–Tully descaled .scups effective collision strengths); only Fe I (1-D) and the density-dependent Fe II statistical-equilibrium coefficient (2-D) remain tabulated. Throughout, $\Lambda(T)$ denotes the effective cooling coefficient *per* ($n_e n_{\text{ion}}$) in $\text{erg cm}^3 \text{s}^{-1}$, in the optically thin limit (every collisional excitation is followed by a radiative decay); the cooling assembly multiplies by $\beta_{\text{esc}} n_e n_{\text{ion}}$. Lower levels are populated in Boltzmann equilibrium over the ground-term fine structure where one exists. Two practical conventions matter when reproducing the numbers: (i) the fits and every number quoted here were made with the rounded $k_B = 1.38 \times 10^{-16} \text{erg K}^{-1}$, the code-wide value until 2026-08-15; the code now carries the CODATA 1.380649×10^{-16} (Update_EXHALE §62.2), a 4.7×10^{-4} relative shift that is negligible against the fit errors; (ii) the transition energies recovered by the fits are the CHIANTI .scups *theoretical* energies, which can differ from the observed-wavelength energies by a few percent (e.g. Mg II 4.277 vs. 4.43 eV) — using the observed value inflates the fit error from $\sim 1\%$ to $\sim 5\%$.

Fit scripts (in `cooling_data/`): `fit_cooling_formulas.py` (resonance lines, Fe II), `fit_cno_formulas.py` (C/N/O), `fit_fs_saturation.py` (fine-structure saturation). The companion notebooks `cooling_formula_fits.ipynb` and `cno_cooling_comparison.ipynb` reproduce every figure here.

2. Resonance lines: exact two-level skeleton

A single-channel resonance line obeys exactly

$$\Lambda(T) = \frac{8.629 \times 10^{-6}}{g_l \sqrt{T}} \Upsilon(T) \Delta E e^{-\Delta E/k_B T}, \quad (1)$$

so the only quantity to fit is the slowly varying effective collision strength, for which the Burgess–Tully type-1 asymptotic form

$$\Upsilon(T) = a + b \ln(1 + T/T_0) \quad (2)$$

suffices. Table 1 lists the fitted constants; Fig. 1 compares the formulas with the source tables. Na I D is reproduced essentially exactly because the underlying rate is itself a Van Regemorter expression with constant Υ (calibrated to CHIANTI Mg II/Ca II with $\bar{g} \simeq 0.2$; Na I has no CHIANTI electron-collision model).

Table 1: Resonance-line cooling: Eqs. (1–2). ΔE is the fitted (.scups theoretical) transition energy. “Max err” is the maximum deviation from the 201-point CHIANTI source table over 10^3 – 10^5 K.

channel	g_l	ΔE [eV]	a	b	T_0 [K]	max err
Mg I $\lambda 2853$	1	4.3381	0.355792	15.1223	94126.5	2.8%
Mg II h&k	2	4.2766	16.2305	20.3448	121896	1.1%
Ca II H&K	2	3.1438	14.7415	19.4920	36218.3	1.2%
Na I D	2	2.1037	36.0744	0 (const.)	—	0.01%

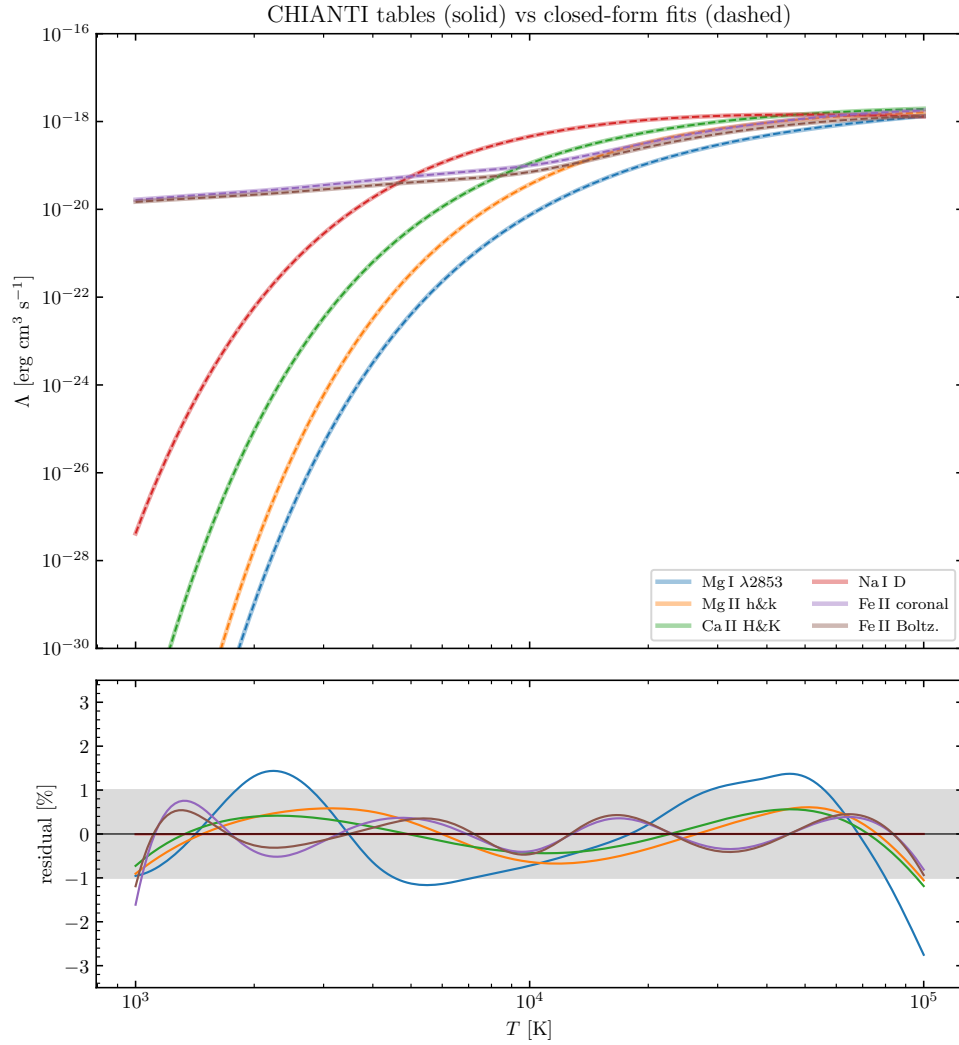


Figure 1: CHIANTI cooling tables (solid) vs. the closed-form fits (dashed) for the four resonance channels and the two Fe II population models (Sect. 3.). Bottom: signed residuals; the grey band is $\pm 1\%$.

3. Fe II: sum of effective two-level groups

Fe II cooling sums hundreds of transitions with ΔE from ~ 0.05 eV (a^6D fine structure) to several eV, so a single exponential cannot represent it. Four effective two-level groups do:

$$\Lambda(T) = T^{-1/2} \sum_{i=1}^4 A_i e^{-T_i/T}. \quad (3)$$

The fitted T_i (Table 2) map onto physically meaningful transition groups — infrared fine structure, optical metastable, UV resonance, and a high-energy tail (Fig. 2).

Table 2: Fe II coefficients for Eq. (3). “Coronal” populates the ground level only; “Boltzmann” distributes the lower levels over the metastable manifold ($E < 4.8$ eV, Huang 2023).

i	Fe II coronal (max err 1.6%)		Fe II Boltzmann (max err 1.2%)	
	A_i	T_i [K]	A_i	T_i [K]
1	1.9889×10^{-18}	1350.1	1.5019×10^{-18}	1175.7
2	1.6793×10^{-17}	9630.3	9.7419×10^{-18}	8678.5
3	6.5362×10^{-16}	58939.3	5.2485×10^{-16}	57594.7
4	7.4638×10^{-16}	139593.3	4.7387×10^{-16}	140620.2

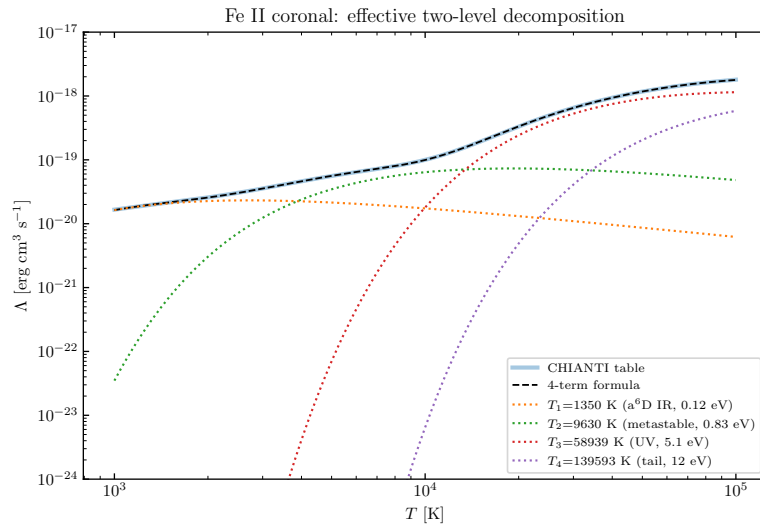


Figure 2: Fe II coronal cooling decomposed into the four effective two-level groups of Eq. (3).

In the cooling *assembly* the coronal formula is superseded by the 2-D statistical-equilibrium table $\Lambda_{\text{eff}}(T, n_e)$ (cool_logL_FeII_ne): the forbidden a^6D lines have $n_{\text{crit}} \sim 10^4$ – 10^7 cm $^{-3}$ and saturate at the dense base. The closed-form coronal expression is retained for the post-process scalar fallback and matches the low- n_e edge of the 2-D table to $\sim 1.6\%$.

4. C, N, O: CHIANTI fits vs. the AIOLOS formulas

The C/N/O coolants inherited from AIOLOS (photochem.cpp; no in-code citation, likely Cloudy-calibrated) are

$$\begin{aligned}\Lambda_{\text{CI}}^{\text{AIO}} &= 10^{-24} + 3.1 \times 10^{-20} e^{-15162/T} [1 + (T/2 \times 10^4)^{1.5}], \\ \Lambda_{\text{CII}}^{\text{AIO}} &= 1.5 \times 10^{-23} + 3.1 \times 10^{-20} e^{-45162/T} [1 + (T/7500)^{1.5}], \\ \Lambda_{\text{OI}}^{\text{AIO}} &= 5.5 \times 10^{-24} + 1.1 \times 10^{-20} e^{-30162/T} [1 + (T/7500)^{0.5}], \\ \Lambda_{\text{OII}}^{\text{AIO}} &= 5.1 \times 10^{-20} e^{-35162/T} [1 + (T/7500)^{0.5}],\end{aligned}\tag{4}$$

with no nitrogen cooling at all. The replacement CHIANTI fits use the same multi-exponential form as Fe II, Eq. (3) with 4–6 terms (Table 3); lower levels are Boltzmann-populated over the ground-term fine structure (C I ³P, C II ²P, N II ³P, O I ³P; the ⁴S ground of N I and O II is a single level).

Figure 3 compares the three. The AIOLOS carbon fits track CHIANTI remarkably well near 10⁴ K (C I to ~1%), which suggests they were calibrated against data of similar provenance; the oxygen fits however deviate by 40–70% across the wind-launch region (1–3 × 10⁴ K), and nitrogen — a non-negligible coolant around 2 × 10⁴ K — was absent entirely. The runtime key `cno_cool` in `metals.inp` selects the source (1 = CHIANTI fits, the default; 0 = AIOLOS).

Table 3: C/N/O coefficients for Eq. (3) (CHIANTI ground-term curves; per ($n_e n_{\text{ion}}$)). The notation $a(b)$ means $a \times 10^b$. Max err is vs. the recomputed CHIANTI curve over 10³–10⁵ K.

ion (max err)		$i=1$	$i=2$	$i=3$	$i=4$	$i=5$	$i=6$
C I (1.8%)	A_i	1.106(−20)	1.392(−18)	6.962(−18)	6.593(−17)	3.017(−16)	
	T_i [K]	2351	15172	25411	72609	177264	
C II (0.5%)	A_i	3.004(−20)	2.410(−20)	3.904(−17)	3.332(−16)	1.360(−15)	
	T_i [K]	61.4	7183	64104	124809	246136	
N I (0.6%)	A_i	1.514(−18)	4.877(−18)	1.160(−17)	1.575(−16)	3.175(−16)	
	T_i [K]	29686	34694	50741	139916	254192	
N II (0.1%)	A_i	2.406(−20)	1.244(−20)	8.040(−18)	1.545(−17)	2.884(−16)	6.944(−16)
	T_i [K]	49.9	6521	23771	64487	157103	295894
O I (2.1%)	A_i	3.236(−22)	6.572(−20)	1.803(−18)	6.762(−18)	3.372(−17)	
	T_i [K]	848	19059	31568	70528	178827	
O II (0.2%)	A_i	1.768(−17)	8.071(−18)	2.344(−16)	8.789(−16)		
	T_i [K]	44129	59732	179091	325669		

5. Ground-term fine-structure statistical equilibrium

The coronal C/N/O curves carry the ground-term fine-structure (FS) lines in the optically thin, *low-density* limit, although their critical densities are of order 10⁰–10⁵ cm^{−3}, decades below the base density of an irradiated atmosphere ($n_e \sim 10^9$, $n_{\text{HI}} \sim 10^{14}$ cm^{−3}). At the converged base of the four planet runs the coronal form overshoots what the same

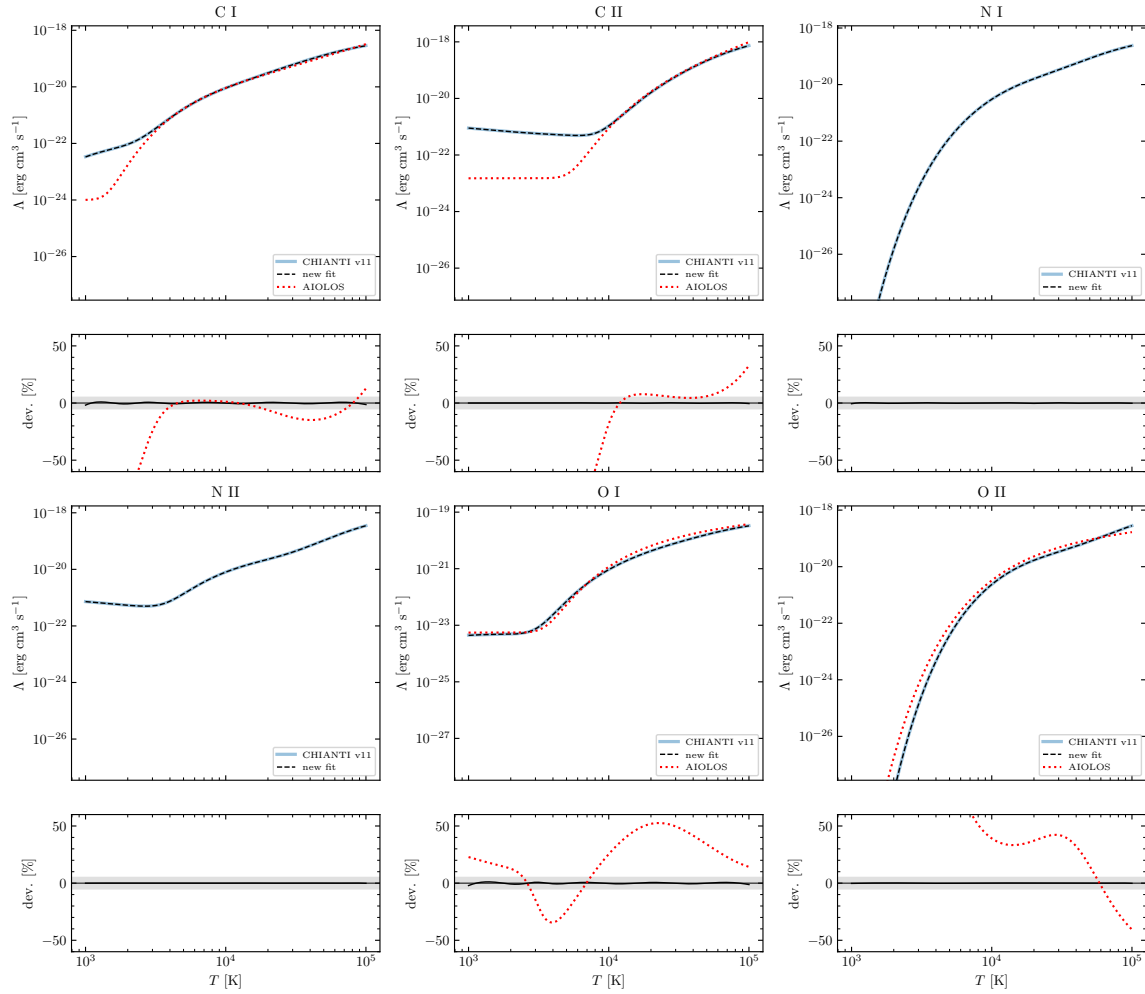


Figure 3: C/N/O line cooling: CHIANTI v11 (solid blue), the new closed-form fits (black dashed), and the AIOLOS formulas (red dotted; absent for nitrogen). Deviation strips show the percentage difference from CHIANTI; the grey band is $\pm 5\%$.

lines can actually radiate by 2.9–8.3 decades (Table 4). In the CHIANTI mode the FS part is therefore replaced, for every C/N/O coolant whose ground term is split, by the exact statistical-equilibrium (SE) solution of that term,

$$\sum_j f_j R_{ji} = f_i \sum_j R_{ij}, \quad \sum_i f_i = 1, \quad (5)$$

$$R_{ul} = C_{ul} + \beta_{ul} A_{ul}, \quad R_{lu} = C_{ul} \frac{g_u}{g_l} e^{-E_{ul}/k_B T}, \quad C_{ul} = n_e k_{e,ul}(T) + n_{\text{HI}} k_{\text{H},ul}(T), \quad (6)$$

$$W_{\text{FS}} = \sum_{u>l} f_u \beta_{ul} A_{ul} k_B E_{ul} \quad [\text{erg s}^{-1} \text{ per ion}], \quad (7)$$

together with a multi-exponential refit $\Lambda_{\text{rem}}(T)$ of the CHIANTI curve that keeps *only* the channels leaving the ground term. The assembly uses $\Lambda_{\text{eff}} = W_{\text{FS}}/n_e + \Lambda_{\text{rem}}(T)$, so the n_e prefactor cancels and the H-collision channel — which the electron-only coronal curves lack entirely — is recovered exactly. The ions treated are the complete set of split ground terms among the coolants:

ion	ground term	levels	lines
C I	$2p^2 \ ^3P_{0,1,2}$	3	[C I] 609.1, 370.4 μm
C II	$2p \ ^2P_{1/2,3/2}$	2	[C II] 157.7 μm
N II	$2p^2 \ ^3P_{0,1,2}$	3	[N II] 205.3, 121.8 μm
O I	$2p^4 \ ^3P_{2,1,0}$	3	[O I] 63.2, 145.5, 44.1 μm

N I and O II have a single-level 4S ground term; Mg I/II, Ca II and Na I have a single ground level; Fe I is built from permitted lines only; and Fe II is already density-dependent through its two-dimensional SE table (Section 3.). The $^3P_2\text{--}^3P_0$ channels of C I and N II have no CHIANTI transition probability (magnetic quadrupole, $A < 10^{-12} \text{ s}^{-1}$) and enter as collisional couplings only.

Limits. As $n_e, n_{\text{HI}} \rightarrow 0$ the SE populations collapse onto the ground level and Eq. (7) reproduces the coronal within-term channel to the accuracy of the Υ fits (1.3–4.4%). As either density $\rightarrow \infty$ it saturates at the exact multilevel LTE emission $\sum_u f_u^{\text{Boltz}} A_{ul} h\nu_{ul}$, verified to 10^{-14} relative. The earlier two-level treatment of [O I] 63 μm and [C II] 158 μm multiplied a two-level solution — already normalized to the *two-level* partition function — by the ground-term Boltzmann fraction of its lower level; that double normalization left its LTE limit low by a factor $1 + (g_u/g_l)e^{-E/k_B T}$, i.e. $2.7\times$ for [C II] 158 μm at the 530 K base of HD 189733 b. Λ_{rem} keeps the ground-term Boltzmann weighting of its lower levels, the convention of the CHIANTI curves it is fitted to; that is the right weighting wherever it matters, and where it is not the ground-only and Boltzmann weightings agree to 1.4–1.9% anyway, because $\Upsilon_{lu} \propto g_l$ for the LS-coupled transitions that leave the ground term.

Line trapping. β_{ul} is the escape probability of that FS line and enters as $A_{ul} \rightarrow \beta_{ul} A_{ul}$ inside Eq. (5), not as a factor on Eq. (7). That is the physically correct place: in the subcritical limit the cooling is set by the collisional excitation rate and is independent of β , while in the saturated limit the escaping flux is proportional to β . The functional form is the Doppler-core single-flight escape probability of Hollenbach & McKee (1979) and de Jong, Boland & Dalgarno

(1980), per face,

$$\beta_1(\tau) = \begin{cases} \frac{1 - e^{-a\tau}}{2a\tau}, & \tau < \tau_c, \\ \frac{1}{4\tau\sqrt{\ln(\tau/\sqrt{\pi})}}, & \tau \geq \tau_c, \end{cases} \quad a = 2.34, \quad (8)$$

with τ the line-center optical depth from the cell to that face; the two branches meet continuously in value at $\tau_c = \sqrt{\pi} e^{a^2/4} \simeq 6.97$ (the residual slope kink there is below 1%), and $\beta_1(0) = 1/2$ — half of the photons leave through the near face. The escape probability of a cell is the two-face sum, $\beta_{ul} = \beta_1(\tau_{up}) + \beta_1(\tau_{dn})$ (`line_escape_probability_one_face`, `Cool_coeff.f90`); with Base IR field on, the downward face also carries the blackbody photon occupation number of the atmosphere below (`docs/lower_atmosphere_coupling`). The opacity of each line uses the same atomic data and the ground-term Boltzmann lower-level population. Λ_{rem} is left optically thin.

Atomic data. Levels, A values and the electron collision strengths are CHIANTI v11.0.2 (`elvlc / wgfa / scups`); $\log_{10} \Upsilon$ is a quartic in $x = \log_{10}(T/10^4)$ fitted over 10^3 – 10^5 K (max error 4.8%, against 8.8% for a cubic on the C I 3P_0 – 3P_2 strength) and T is clamped to that range. The electron de-excitation rate is $k_e = 8.629 \times 10^{-6} \Upsilon(T)/(g_u \sqrt{T})$. The H-atom de-excitation rates are taken from quantum scattering calculations, $\log_{10} k_H$ a quadratic in $\log_{10}(T/10^3)$ with T clamped to the tabulated range (max error 4.9%):

ion	source	range
C I, N II	Yan & Babb (2023), MNRAS 518, 6004, Tables 1–2	10 – 10^4 K
O I	Abrahamsson, Krems & Dalgarno (2007), ApJ 654, 1171	20 – 1000 K
C II	Barinovs et al. (2005), ApJ 620, 537	20 – 2000 K

These replace the values of the legacy `use_2lev_cool` branch (a constant 4.0×10^{-11} for [C II] $158 \mu\text{m} + \text{H}$ and $4.2 \times 10^{-11}(T/100)^{0.67}$ for [O I] $63 \mu\text{m} + \text{H}$, both flagged approximate), which are 5–20 $\times$ below them. H collisions on the *ion* N II are included on the same footing as on the neutrals; they are not what saturates N II at the base ($n_{\text{crit,e}}$ of [N II] $205 \mu\text{m}$ is $\sim 10^2 \text{cm}^{-3}$), they matter only in gas thin enough for the electron channel alone to be subcritical. `cooling_data/fit_fs_saturation.py` is the auditable source and prints every coefficient.

Impact. The effect is confined to the base but is not small there. On HD 189733 b the C I coronal floor had been the dominant coolant of the whole layer below $r \simeq 1.01 R_p$ (3.3×10^{-6} against a local heating of $3.4 \times 10^{-6} \text{erg cm}^{-3} \text{s}^{-1}$); with the ground term saturated it falls to 10^{-11} , the base warms from 528 to 613 K, the density at $1.01 R_p$ rises by a factor 3.3 and the mass-loss rate at $r = 2 R_p$ increases by 21%, from 2.35×10^9 to $2.85 \times 10^9 \text{g s}^{-1}$. Outside $r \simeq 1.1 R_p$ the temperature profile moves by less than 4%. The same change also removes the sensitivity of the base to the coronal-fit cutoff width w : see `docs/coronal_cutoff_width.md`.

6. Free–free and the remaining tabulated pieces

Free–free cooling keeps the Huang (2023) Eq. (9) form, $\Lambda_{\text{ff}} = 1.9095 \times 10^{-25} Z^2 (T/10^4)^{0.55}$ per $(n_e n_{\text{ion}})$. Two coefficients remain tabulated because they are not reducible to closed form: Fe I (NIST f -values + Van Regemorter over the

Table 4: Ground-term fine-structure cooling at the converged base cell of each run: the coronal within-term channel as the CHIANTI fits carry it, against the exact SE value at the local (n_e, n_{HI}) , both volumetric [$\text{erg cm}^{-3} \text{s}^{-1}$], $\beta = 1$. The last column is the SE value as a fraction of the local heating rate.

run	ion	coronal	SE (saturated)	decades	SE/heat
HD 189733 b	C I	1.61×10^{-4}	2.00×10^{-11}	6.9	5.9×10^{-6}
$T = 528 \text{ K}$	C II	1.14×10^{-4}	2.09×10^{-12}	7.7	6.2×10^{-7}
	N II	7.47×10^{-10}	5.97×10^{-17}	7.1	1.8×10^{-11}
	O I	1.47×10^{-4}	3.08×10^{-8}	3.7	9.1×10^{-3}
HD 209458 b	C I	7.77×10^{-5}	5.76×10^{-11}	6.1	3.0×10^{-4}
$T = 519 \text{ K}$	C II	2.50×10^{-12}	2.56×10^{-19}	7.0	1.3×10^{-12}
	N II	1.28×10^{-15}	5.72×10^{-22}	6.3	3.0×10^{-15}
	O I	7.42×10^{-5}	8.77×10^{-8}	2.9	4.5×10^{-1}
WASP-52 b	C I	2.15×10^{-5}	1.94×10^{-12}	7.0	1.1×10^{-6}
$T = 794 \text{ K}$	C II	6.14×10^{-5}	2.43×10^{-12}	7.4	1.4×10^{-6}
	N II	8.42×10^{-10}	1.48×10^{-16}	6.8	8.2×10^{-11}
	O I	8.92×10^{-6}	3.40×10^{-9}	3.4	1.9×10^{-3}
WASP-121 b	C I	8.08×10^{-6}	4.48×10^{-14}	8.3	5.2×10^{-9}
$T = 2407 \text{ K}$	C II	8.19×10^{-4}	1.33×10^{-11}	7.8	1.6×10^{-6}
	N II	1.14×10^{-8}	8.49×10^{-16}	7.1	9.9×10^{-11}
	O I	1.26×10^{-5}	1.20×10^{-9}	4.0	1.4×10^{-4}

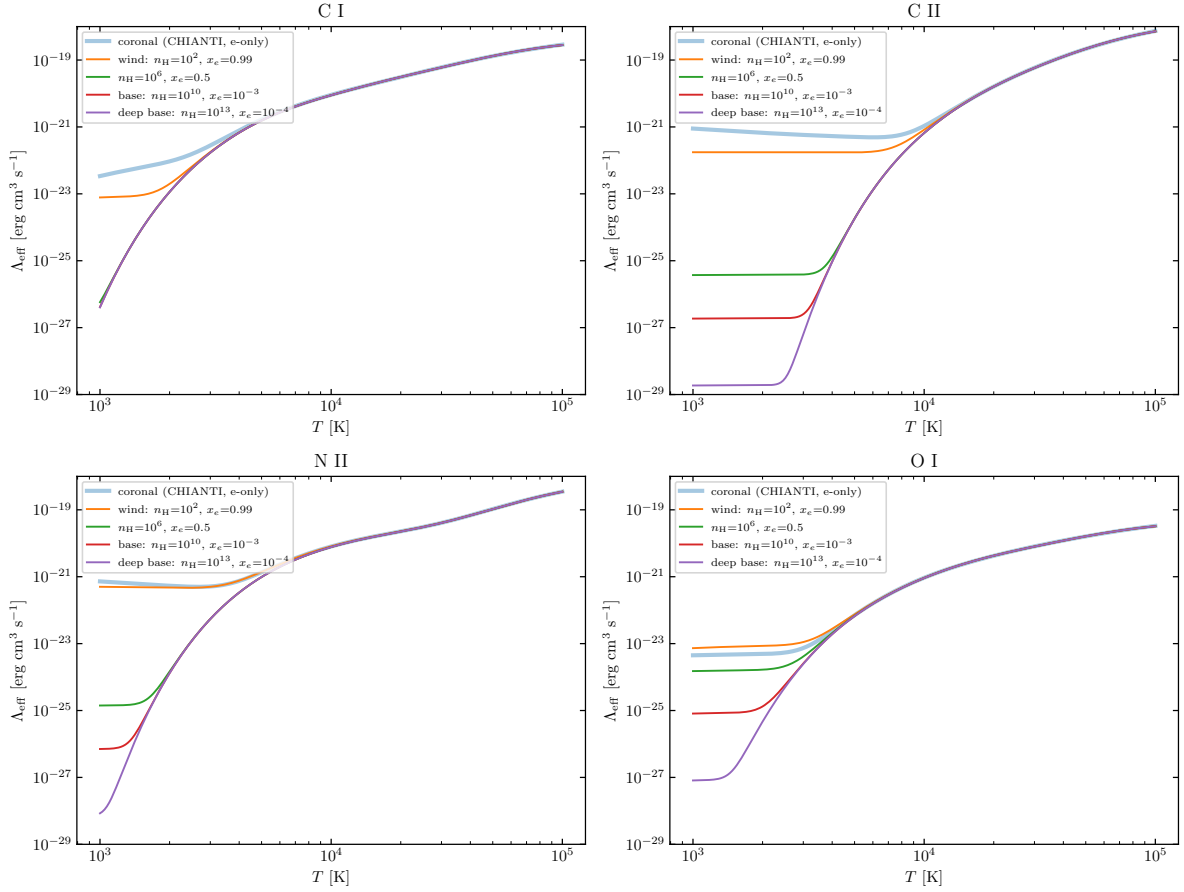


Figure 4: Effective CI, CII, NII and OI cooling coefficients with the ground term solved in statistical equilibrium, Eqs. (5) and (7), for densities from the wind to the deep base, compared with the electron-only coronal curve. The low-temperature plateau of each coronal curve is the unsaturated fine-structure floor; it collapses by several decades once the ground term is allowed to thermalise.

even-parity metastable manifold; 1-D in T) and the Fe II multilevel statistical-equilibrium coefficient $\Lambda_{\text{eff}}(T, n_e)$ (2-D). A possible future step, analogous to Sect. 5., would be a closed-form saturation treatment of the Fe II a^6D manifold, but the 2-D table is exact within its grid and carries no runtime cost concern.

Known limitations. (i) The legacy `use_2lev_cool` branch hard-codes the AIOLOS exponential parts for C II/O I and is not reached by the CHIANTI formulas; it is superseded by Sect. 5. and kept only for comparison (default off). (ii) The H-collision rates of Sect. 5. are fits to published quantum scattering calculations, but O I is tabulated only to 1000 K and C II to 2000 K; above those the rate is held at its endpoint value rather than extrapolated. (iii) All formulas are calibrated over 10^3 – 10^5 K; outside this range they extrapolate smoothly but are unvalidated.